

ZACHARY LIEBERMAN

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Education

University of Kansas

Expected May 2029

Bachelor of Science in Mechanical Engineering

- **GPA:** 3.73 / 4.00

Skills

CAD / Design: SolidWorks, AutoCAD, Onshape, Fusion360, Blender

Fabrication / Prototyping : 3D and 2D Modeling, Rapid Prototyping, Soldering, Hand and Power Tools

Programming / Software: JavaScript, Node.js, HTML/CSS, Typescript, Python, Git

Experience

DSI Club

August 2024 – May 2025

Lead Staff Member

Deerfield, Illinois

- Developed external commercial broadcast partnerships with organizations including Marquee Sports Network, creating expanded televised showcase opportunities for student productions.
- Managed and mentored a student crew of camera operators and on-air staff, optimizing live broadcast workflows and maintaining high production standards.
- Coordinated production logistics and technical equipment setup for multi-sport live broadcasts, ensuring seamless execution across seasonal high school athletic tournaments.

Black Squirrel Timing Company

August 2025 – Present

Track and Cross Country Meet Timer

Lawrence, Kansas

- Implemented electronic timing systems and published live results across multi-state events, achieving sub-second data accuracy under strict deadlines.
- Championed the transition to newer timing software, improving overall system reliability and race day efficiency.

PFv Performance LLC

February 2024 – May 2025

Track and Cross Country Meet Timer

Deerfield, Illinois

- Automated finish-line video capture and cloud uploads to Athletic.net by developing custom software in JavaScript, generating roughly \$3,000 in upside revenue at \$4 per race.
- Decreased system downtime during live events by architecting remote diagnostics protocols and managing field equipment maintenance.

Projects

FTC Robotics Robot | Onshape, Blender, Android Studio, VS Code

April 2024

- Designed 3D CAD assemblies and functional mechanism animations in Onshape and Blender, securing technical judging points to help win the 2025 Illinois State Championship and finish Top 50 at Worlds.
- Calibrated onboard sensor arrays and actuator feedback loops in collaboration with the drive team in Android Studio, optimizing autonomous scoring consistency and field positioning.
- Diagnosed and repaired electromechanical subsystem failures during high-pressure competition rounds, ensuring 100% field uptime through championship eliminations.

Scrapescraper.live | JavaScript, Node.js, HTML/CSS

August 2024

- Architected and deployed a full-stack live sports tracking platform that scrapes and parses real-time game data across high school, collegiate, and professional tiers.
- Designed a responsive web interface to display dynamic scores and event statistics on video scoreboards, improving fan engagement across multiple regional high schools and colleges, including the University of Kansas.
- Optimized automated data scraping pipelines and server workflows in Node.js, reducing live latency and maintaining reliable data delivery during peak multi-game traffic hours.

Leadership and Awards

- School of Engineering Dean's Honor Roll (Fall 2025)
- Vanguard Interschool Engineering, V.I.S.E
- American Society of Mechanical Engineers, ASME